UNIT 3 · TOPIC 2(C)

Testing of Seed Drills and Planters

Syllabus topic: Methods of testing and performance evaluation of tillage equipment, seed drills and planters

Laboratory tests (RNAM/BIS — e.g., IS 6316)

- Calibration and seed-rate accuracy at the recommended settings (Topic 1).
- **Uniformity between rows** — CV of per-opener collections.
- Rate stability versus **hopper fill level** (full/half/quarter), **machine tilt** (fore-aft and lateral slopes simulating the field) and operating speed.
- **Visible mechanical seed damage** — sample before/after metering (BIS limit ~**0.5%** for cereals) — and **germination** of metered seed.

Seed distribution along the row — greased/sticky belt test

1. Metered seeds fall on a moving greased belt.
2. Measure the spacings.
3. Compute the miss index, multiple index, quality-of-feed index and precision (CV) — see Topic 1b for the definitions.

Field tests

- Depth of seed placement and its uniformity (careful excavation); row-spacing check.
- Plant emergence count and **plant stand vs target population**; skips/doubles in the field.
- Draft & power; field capacity/efficiency; hill placement for hill-drop planters.
- Performance across the recommended speed range (metering deteriorates at high speed).

Practice MCQs

Q49. Seed-drill rate stability must be checked at different:

(a) Hopper fill levels and machine tilts (b) Paint colours
(c) Tyre brands (d) Hitch pins

Q50. Uniformity of seed spacing along the row is tested in the lab on a:

(a) Patternator (b) Greased/sticky belt
(c) Soil bin (d) Test track

Q51. Permissible visible mechanical seed damage in drill metering (BIS, cereals) is about:

(a) 0.5% (b) 2%
(c) 5% (d) 10%

Q52. Metering uniformity between rows is expressed by:

(a) Mean only (b) Coefficient of variation of per-opener collections
(c) Mode (d) Range only

Q53. Germination test of metered seed detects:

(a) Visible damage only (b) Internal (invisible) mechanical damage
(c) Seed colour (d) Purity

Q54. Field evaluation of a planter compares plant stand with:

(a) Fuel used
(b) Target population
(c) Draft
(d) Rainfall

Answers

Q49 — (a) Hopper fill levels and machine tilts

Q50 — (b) Greased/sticky belt

Q51 — (a) 0.5% — *Visible damage is counted by inspection; germination testing exposes internal cracks that inspection misses (cf. combine seed damage, Unit 2 Topic 8).*

Q52 — (b) Coefficient of variation of per-opener collections

Q53 — (b) Internal (invisible) mechanical damage — *Germination testing exposes internal cracks that visual inspection misses.*

Q54 — (b) Target population

Sources: RNAM (UN-ESCAP) Test Codes and Procedures for Farm Machinery; BIS Indian Standards for seed drills, sprayers and dusters; Kepner et al. (performance concepts); Jagdishwar Sahay; FMTTI (Budni) test procedures; NPTEL lectures.
